

Simulation Object Documentation Module (SODM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Operational Layer: Simulation Object Governance Layer

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Governed Space: Simulation Object Documentation

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Canonical Language: English (UCL™)

Governed Space

Simulation Object Documentation

Canonical Definition

Simulation Object Documentation Module (SODM) defines the governance conditions under which the identity, boundary, version, configuration, material characteristics, changes, limitations, and simulation-relevant state of a Simulation Object are consolidated into a persistent, traceable, reconstructable, and operationally usable governed record.

SODM ensures that the Simulation Object established through the preceding SOS modules does not remain distributed across disconnected definitions, assumptions, records, systems, or undocumented knowledge.

It creates the persistent documentary representation through which the exact object intended for simulation can be reconstructed and referenced throughout the SIMULOS® lifecycle.

Operational Role

SODM serves as the fifth and final internal governance space of the Simulation Object Standard (SOS).

The preceding modules establish:

SOIM — What exactly is the Simulation Object?

SOBM — Where does the Simulation Object begin and end?

SOVM — Which materially distinguishable version of the Simulation Object is being simulated?

SOCM — What exact configuration of that version is being simulated?

SODM asks:

How must the complete governed definition of that Simulation Object be documented and preserved so that every downstream simulation activity refers to the same reconstructable object basis?

The governed progression is:

Object Identity → Object Boundary → Object Version → Object Configuration → Documentation Basis → Object Record Formation → Documentation Qualification → Simulation Object Reference → Documentation Change → Persistent Simulation Object Record

Module Operational Space

SODM governs: Simulation Object documentation basis, object record formation, object definition consolidation, object identity documentation, object boundary documentation, version documentation, configuration documentation, material characteristic documentation, inclusion documentation, exclusion documentation, object-state documentation, object limitation documentation, object dependency references, object relationship references, object documentation completeness, object documentation consistency, object documentation currency, object documentation ambiguity, object documentation conflict, object documentation gaps, object documentation change, object documentation supersession, object record status, simulation-object reference, downstream object referencing, point-in-time reconstruction, historical object documentation, documentation continuity, and Simulation Object record traceability.

Module Function

A Simulation Object may be correctly identified, bounded, versioned, and configured while still remaining operationally unusable for governed simulation if those characteristics cannot be reconstructed from a coherent record.

Material information may otherwise exist across: engineering documents, software repositories, configuration systems, model inventories, technical specifications, databases, operational records, human knowledge, external documentation, or temporary simulation records.

SODM prevents: undocumented Simulation Objects, fragmented object definitions, downstream use of inconsistent object descriptions, loss of object boundaries, undocumented version or configuration state, obsolete object records being treated as current, silent modification of the documented object, historical object states being overwritten, object limitations disappearing from

downstream simulation activity, and simulation findings becoming detached from the object definition on which they were based.

Simulation Object Documentation Basis

The Simulation Object Documentation Basis is the governed set of information required to establish a sufficiently complete documentary representation of the Simulation Object.

It may draw from: SOS governance records, technical documentation, controlled object records, configuration records, authoritative specifications, system descriptions, engineering records, object inventories, operational records, or other identifiable sources.

SODM does not require all information to exist in one technical system.

It requires the governed Simulation Object definition to remain reconstructable.

Simulation Object Record

A Simulation Object Record is the persistent governed representation of the Simulation Object used throughout the SIMULOS® lifecycle.

At minimum, where applicable, it should establish: Simulation Object Identifier, object name or designation, object type, object description, object boundary, material inclusions, material exclusions, object version, object configuration, material characteristics, relevant operational state, known limitations, material dependencies, relevant relationships, documentation status, effective point, historical references, and sufficient traceability to the underlying SOS governance basis.

Documentation Consolidation

SODM does not require duplication of every underlying document.

Instead, it establishes a governed documentary layer capable of connecting distributed object information into one coherent Simulation Object definition.

Conceptually:

Identity Record + Boundary Record + Version Record + Configuration Record + Material Object Information

↓

Simulation Object Record

The record may therefore operate as a controlled reference layer rather than a duplicate technical repository.

Authoritative Object Reference

A governed simulation should have an identifiable Authoritative Simulation Object Reference.

This reference establishes which Simulation Object Record governs the simulation.

Where multiple object descriptions exist, the authoritative reference prevents downstream simulation activities from selecting incompatible object definitions without visibility.

Documentation Completeness

Documentation Completeness asks:

Is sufficient information available to reconstruct the Simulation Object relevant to the intended simulation?

Completeness is relative to the Simulation Object and the material information required to distinguish it.

It does not require documentation of every technically knowable characteristic.

Material Documentation Requirement

A characteristic should be documented where its omission could materially affect: object identification, object distinction, object boundary, version identification, configuration identification, downstream simulation design, interpretation of simulation results, or later reconstruction of what was simulated.

Documentation Consistency

Documentation Consistency asks whether material Simulation Object records describe a mutually compatible object state.

For example:

Version Record: V4.2

and:

Configuration Record: only valid for V5.0

would create a documentation inconsistency requiring resolution.

Documentation Currency

A Simulation Object Record should indicate the object state and effective period it represents.

A record may remain historically valid while no longer being current.

Therefore:

Historically Accurate ≠ Currently

Applicable Current Documentation State

Where the Simulation Object changes, SODM should make it possible to distinguish:

Current Object Record

from:

Historical Object Record

and:

Superseded Object Record

This prevents documentation history from being mistaken for present object reality.

Documentation Ambiguity

Documentation Ambiguity exists where the available records permit multiple materially different interpretations of the Simulation Object.

Examples include uncertainty regarding: object identity, boundary, version, configuration, included components, excluded components, or relevant operational state.

Material ambiguity should remain explicit until resolved.

Documentation Conflict

A Documentation Conflict exists where credible object records materially contradict one another.

For example:

Record A → Configuration C4

Record B → Configuration C7

for the same relevant object state.

SODM requires the conflict to remain visible.

It must not be silently resolved through undocumented preference.

Documentation Gap

A Documentation Gap exists where information necessary to reconstruct a material characteristic of the Simulation Object is absent.

Possible statuses may include:

Non-Material Gap

Conditionally Acceptable Gap

Material Gap

or:

Gap Materiality Undetermined

The gap itself does not determine simulation validity.

It becomes an explicit input to downstream simulation governance.

Documentation Qualification

Before the Simulation Object Record becomes the governing object reference, its documentation basis should be sufficiently qualified for: completeness, consistency, currency, applicability, ambiguity, conflict, and material gaps.

SODM does not perform general evidence validation.

It determines whether the Simulation Object has a sufficiently usable governed documentary representation.

No Undocumented Object Substitution Rule

SIMULOS® establishes:

A Simulation Object materially different from the documented governing object must not be substituted without explicit change to the Simulation Object Record.

The simulation cannot silently move from:

Object A

to:

Object A'

while preserving the original object reference.

No Documentation Override Rule

An informal description, label, filename, or operational shorthand must not silently override a materially different governed Simulation Object Record.

Where disagreement exists, the difference should be resolved or preserved as a conflict.

Documentation Change

A Documentation Change occurs when information within the Simulation Object Record is modified.

A documentation change may represent: correction, clarification, additional information, object change, version change, configuration change, boundary change, status change, or removal of obsolete information.

The reason for material changes should remain traceable. Documentation Change ≠ Object Change

SIMULOS® distinguishes:

Documentation Change

from:

Simulation Object Change

Correcting an incorrectly recorded serial number may change documentation without changing the object.

Changing the physical subsystem represented may change the object itself.

SODM records the documentary consequence.

The appropriate preceding SOS module determines the underlying object-governance consequence.

Material Documentation Change

A documentation change is material where it changes how the Simulation Object would be: identified, bounded, versioned, configured, interpreted, distinguished, or represented in downstream simulation activity.

Material documentation changes should generate downstream reassessment where necessary.

Documentation Supersession

Where a new Simulation Object Record replaces a previous record, the previous record should not disappear.

The relationship should remain reconstructable:

Previous Record → Supersession Event → Successor Record Historical Preservation

Historical Simulation Object Records should remain available where required to reconstruct previous simulations.

This enables questions such as:

What object definition governed Simulation Run SR-104?

Which version and configuration were documented at that time?

Which boundaries were applicable?

Which limitations were known?

Point-in-Time Reconstruction

SODM should support reconstruction of the Simulation Object as it was documented at a specific relevant point.

This is particularly important where: the object evolves, configuration changes, multiple simulations occur over time, simulation outputs are later reused, incidents occur, or historical simulation conclusions are reviewed.

Object Documentation Continuity

Documentation Continuity preserves the relationship:

Object Identity

↓

Boundary

↓

Version

↓

Configuration

↓

Simulation Object Record

↓

Simulation Activity

↓

Subsequent Object Change

↓

Successor Object Record

This ensures that object evolution does not erase the basis of earlier simulation activity.

Simulation Object Reference

Once sufficiently established, the Simulation Object Record becomes the controlled reference consumed by subsequent SIMULOS® Parent Standards.

Downstream simulation governance should reference the governed object rather than independently redefining it.

Downstream Reference Principle

SIMULOS® establishes:

A downstream simulation activity should not silently redefine the Simulation Object established by SOS.

Where a downstream requirement demonstrates that the object definition is insufficient or incorrect, the issue should return to the appropriate SOS governance space.

Documentation and Purpose & Scope

SODM establishes the object documentation basis.

It does not determine: why the simulation is being performed, what the simulation must evaluate, what reliance is intended, or how broad the simulation inquiry should be.

Those questions belong to the subsequent Simulation Purpose & Scope Standard.

Documentation and Reality Representation

SODM documents the real or conceptual object claimed to be represented.

It does not determine whether a simulation model accurately represents that object.

That belongs to the later Reality Representation governance space.

Documentation and Simulation Model

A Simulation Object Record is not the Simulation Model.

The distinction is:

Simulation Object Record → What is intended to be represented

Simulation Model → How that object is represented for simulation

Confusing these two layers can cause a model description to replace the actual object definition.

Documentation and Environment

SODM governs documentation of the Simulation Object.

It does not govern the complete simulated environment surrounding that object.

Environmental characteristics belong to the relevant SIMULOS® environment governance space.

Documentation and Evidence

A Simulation Object Record may later become part of a validation or audit evidence basis.

SODM does not determine its general evidential fitness.

Where formal validation is required, VALIDOS® governs validation evidence fitness.

Where audit evidence is required, ADIT® governs the relevant audit-evidence lifecycle.

Documentation and Integrity

SODM does not replace INTEGROS®.

INTEGROS® may govern the integrity of: records, data, documentation structures, storage, change history, or other integrity-relevant elements.

SODM governs specifically whether the Simulation Object possesses a sufficiently persistent and reconstructable documentary representation for simulation governance.

Documentation and Identity

SODM does not replace broader identity governance.

Where identity, origin, ownership, provenance, or transfer require independent governance, complementary OOF® identity architectures may apply.

SODM consumes the identity necessary to maintain the Simulation Object as a stable simulation reference.

Documentation and Auditability

SODM creates traceability necessary for later reconstruction.

It does not govern the complete audit lifecycle.

ADIT® may consume SODM records where continuous or retrospective audit is required.

Documentation Change Propagation

A material change to the Simulation Object Record may affect downstream governance.

Potentially affected spaces may include: Simulation Purpose & Scope, Reality

Representation, Assumptions, Model & Environment, Inputs & Data, Scenario Design, Simulation Execution, Simulation Output, Reality Correlation, Simulation Evidence, Operational Reliance, and Resimulation.

SODM identifies the documentary change.

The affected downstream governance spaces determine its consequence.

Documentation Reassessment Signal

Where a material documentation change may invalidate the current object basis, SODM should generate a Simulation Object Reassessment Signal.

Possible causes include: corrected object identity, changed boundary, new version information, configuration discovery, previously unknown material component, corrected exclusion, newly identified limitation, or resolved documentation conflict.

Documentation Status

A Simulation Object Record may carry a status such as:

Draft

Qualified

Current

Conditionally Current

Superseded

Historical

Conflicted

Incomplete

or:

Undetermined

Status indicates documentary condition.

It does not establish simulation validity.

Persistent Simulation Object Record

The complete SOS lifecycle should result in a persistent record capable of preserving: object identity, object boundary, object version, object configuration, material characteristics, relevant limitations, dependencies, documentation basis, documentation qualification, conflicts, gaps, changes, status, effective period, supersession, historical continuity, and downstream references.

SOS Completion Gate

SODM establishes the final internal completion gate of SOS.

The key question is:

Is the Simulation Object sufficiently identified, bounded, versioned, configured, documented, and reconstructable to enter governed Simulation Purpose & Scope?

Possible outcomes may include:

SOS Complete

SOS Conditionally Complete

SOS Incomplete

or:

SOS Completion Undetermined

A conditional or incomplete status should remain visible downstream.

Minimum Implementation Framework

1. Establish the Documentation Basis

Identify the records necessary to reconstruct the Simulation Object. 2.

Consolidate the Governed Object Definition

Connect identity, boundary, version, configuration, material characteristics, limitations, and relevant references into the Simulation Object Record. 3.

Qualify the Documentation

Assess completeness, consistency, currency, applicability, ambiguity, conflict, and material gaps. 4. Establish the Authoritative Object Reference

Identify the Simulation Object Record that governs downstream simulation activity. 5. Preserve Change and Historical Continuity

Record material documentation changes, supersession, prior object states, and point-in-time applicability. 6. Bind the Record to Downstream Simulation Governance

Ensure subsequent SIMULOS® spaces reference the established Simulation Object rather than silently redefining it. 7. Generate Reassessment Signals

Where new information materially changes the object basis, return the issue to the relevant SOS governance space and propagate the resulting change downstream. 8. Determine SOS Completion

Establish whether the Simulation Object is sufficiently governed to proceed into Simulation Purpose & Scope.

Governance Outputs

SODM may produce: Simulation Object Documentation Basis, Simulation Object Record, Authoritative Simulation Object Reference, Object Definition Consolidation Record, Object Documentation Completeness Assessment, Object Documentation Consistency Assessment, Object Documentation Currency Assessment, Object Documentation Applicability Record, Object Documentation Ambiguity Record, Object Documentation Conflict Record, Object Documentation Gap Record, Material Documentation Change Record, Object Documentation Supersession Record, Historical Simulation Object Record, Point-in-Time Object Reconstruction Record, Object Documentation Status, Simulation Object Reassessment Signal, Downstream Object Reference, Simulation Object Documentation Traceability Map, SOS Completion Record, and Persistent Simulation Object Record.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous vehicle platform is represented across: engineering specifications, sensor configuration records, software version records, and operational configuration databases.

No single record identifies the complete object state used by the simulation.
Application

SODM consolidates the relevant governed references into:

Vehicle AV-17 → Version 6.3 → Configuration C12

The Simulation Object Record identifies the applicable sensor package, control software, material boundaries, exclusions, and known object limitations.
Result

The simulation no longer depends upon an informal assumption about which vehicle configuration is being represented.

Every downstream SIMULOS® space can reference the same governed Simulation Object. Use Case 2 — Spacecraft Simulation History

Scenario

A spacecraft undergoes configuration changes during its operational lifecycle.

A simulation performed two years earlier becomes relevant after an operational anomaly. Application

SODM reconstructs the historical Simulation Object Record applicable to the earlier simulation.

The record establishes:

Object Identity → Historical Version → Historical Configuration → Applicable Boundary → Known Limitations → Simulation Reference Result

Investigators can determine exactly which spacecraft state the historical

simulation represented rather than applying current spacecraft documentation retrospectively.

Architectural Position

Simulation Object Documentation Module (SODM) is the fifth and final internal governance space of the Simulation Object Standard (SOS).

The complete SOS progression is:

Simulation Object Identification → Simulation Object Boundary → Simulation Object Versioning → Simulation Object Configuration → Simulation Object Documentation

Together, these five modules establish the governed Simulation Object before simulation purpose, representation, assumptions, models, data, scenarios, execution, outputs, correlation, evidence, reliance, or lifecycle governance begins.

SOS therefore answers one foundational architectural question:

What exactly is being simulated?

Once SODM establishes a sufficiently persistent and reconstructable Simulation Object Record, the object may proceed to the next SIMULOS® Parent Standard:

Simulation Purpose & Scope Standard.

Foundational Principle

A Simulation Object is not sufficiently governed merely because its identity is known; the complete object basis must remain reconstructable so that every simulation activity can demonstrate exactly what object it represented.

Canonical Closing Statement

Simulation Object Documentation Module (SODM) completes the Simulation Object Standard by consolidating object identity, boundary, version, configuration, material characteristics, limitations, changes, and historical continuity into a persistent and reconstructable Simulation Object Record. Through documentation qualification, authoritative referencing, ambiguity and conflict preservation, change governance, supersession, point-in-time reconstruction, downstream binding, and SOS completion control, SODM prevents simulation activity from becoming detached from the exact object it claims to represent. By establishing a stable documentary bridge between Simulation Object governance and every subsequent SIMULOS® lifecycle space, SODM completes the foundational answer to the first question of Simulation Governance Architecture: exactly what is being simulated.

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